

Task 2 Report: AI Generated Snake Game

Initial Idea: Simple Snake Game

Prompt Used

Create a basic snake game where the snake moves around the screen, eats food, and grows longer.

What Worked Well

- Game did worked
- Easy to understand and test.

Limitations

- Gameplay was repetitive.
- Not mobile friendly
- Game starts on its own
- No difficulty level was added

Iteration 1: Mobile Friendly Snake Game

Prompt Used

Modify the snake game by adding controls on screen. Add various difficulty levels. And change the environment to make it look attractive.

What worked well

- Added on screen controls
- Improved graphics
- Added difficulty levels

Limitations

- The snake movement is way too fast
- The game is too basic

Iteration 2: Educational Snake Game

Prompt Used

Modify the snake game into an educational game for children. Display fruits on the trees and require the player to identify or select the correct fruit before snake make jump from one tree to another.

What worked well

- Added learning objectives.
- Increased user engagement.
- Combined entertainment with education.

Limitations

- It did not work out according to instructions.
- The educational elements did not fully work.
- I am not satisfied with the game it actually does not look like a game

Iteration 3: Snake Adventure: Lost Kingdom (Current Version)

Prompt Request From Chatgpt:

Give me a prompt to

Create a adventurous snake game considering these:

The game should have instructions at start about how to play.

There should be various difficulty levels.

The snake jumps across jungle to cross different hurdles and collect some rewards.

There is a tunnel or end goal where the snake has to reach.

Add voices when snake collects something but don't add any music in the whole game.

The controls should be present on screen.

The game should be both computer and mobile friendly.

It should be easy to deploy

Prompt:

Master Prompt: Snake Adventure – Lost Kingdom

Project Overview

Create a complete HTML5 mobile game called "**Snake Adventure: Lost Kingdom**".

The game should feel like a polished, professional mobile adventure game designed for teenagers and adults. The gameplay should combine classic snake mechanics with exploration, treasure hunting, obstacles, progression systems, and adventure elements.

The final output must be a **single self-contained HTML file** using only HTML, CSS, and JavaScript.

Gameplay Concept

The player controls a snake traveling through a mysterious jungle in search of an ancient lost temple.

The objective is to:

- Explore the jungle.
- Collect coins and treasures.
- Avoid dangers and obstacles.
- Complete increasingly difficult levels.
- Reach the ancient temple at the end of each stage.

The game should feel like a casual adventure game rather than a traditional snake game.

Tutorial & Introduction Screen

Before the game starts:

Intro Video

Display a short animated tutorial or video-style sequence explaining:

- Game objective
- Movement controls
- Jump controls
- Coin collection
- Power-ups
- Health system
- Obstacles

- Checkpoints

Requirements:

- Include a "Skip Tutorial" button.
- Include a "Start Game" button after the tutorial.
- Tutorial should run directly inside the HTML game.
- No external video files.
- Create the tutorial using HTML5 animations, canvas animations, or JavaScript.

Difficulty Selection

Before starting gameplay, allow the player to choose a difficulty level.

Easy

- Slow snake speed
- More health
- More checkpoints
- Fewer obstacles
- More power-ups

Medium

- Balanced gameplay
- Standard health
- Standard obstacles

Hard

- Faster snake speed
- More obstacles
- Less health
- Fewer checkpoints
- More aggressive hunters

Expert

- Very fast gameplay
- Minimal checkpoints
- High obstacle density
- Limited power-ups
- Maximum challenge

The selected difficulty should affect gameplay mechanics throughout the game.

Core Controls

The game must be fully playable on mobile devices.

All controls must be permanently visible on screen.

Mobile Controls

Include:

Direction Controls

- Left button
- Right button

Action Controls

- Jump button
- Pause button

Optional Advanced Controls

- Dash button
- Special ability button (future upgrades)

Requirements:

- Large touch-friendly buttons
- Responsive design
- No keyboard required
- Optimized for portrait orientation

Snake Mechanics

The snake should:

- Move smoothly
- Jump over obstacles
- Collect items
- Take damage
- Use power-ups
- Reach checkpoints
- Complete levels

Animation Requirements:

- Smooth movement
- Head and body animation
- Direction changes
- Jump animation

Level System

Create multiple levels.

Each level should:

- Be longer than the previous one
- Include new obstacles
- Increase difficulty
- Introduce new challenges

Example progression:

Level 1

Training Jungle

Level 2

Deep Forest

Level 3

Ancient Ruins

Level 4

Temple River

Level 5

Lost Kingdom Temple

Obstacles

Include:

Environmental Hazards

- Spikes
- Rivers
- Pits
- Poison plants

Dynamic Hazards

- Falling rocks
- Rolling logs
- Moving traps

Enemies

- Hunters
- Jungle guards
- Temple protectors

Enemies should patrol and create additional challenges.

Collectibles

Coins

Used for:

- Scores
- Unlockables

- Achievements

Treasure Chests

Provide:

- Bonus points
- Coins
- Power-ups

Rare Artifacts

Hidden collectibles that reward exploration.

Power-Up System

Include:

Speed Boost

Temporary speed increase.

Shield

Blocks incoming damage.

Double Coins

Doubles coin rewards.

Magnet

Automatically attracts nearby coins.

Extra Health

Restores health points.

Power-ups should have visible timers.

Health & Lives System

Player should have:

- Health bar
- Damage animations
- Recovery items

Game Over occurs when health reaches zero.

Checkpoint System

Add checkpoints throughout levels.

Requirements:

- Save player progress within the level.
- Respawn from latest checkpoint after death.
- Visual checkpoint indicators.

Graphics & Visual Design

Create a visually appealing jungle world.

Include:

Environment

- Trees
- Plants
- Vines
- Ancient ruins
- Temple structures

Effects

- Particle effects
- Dust effects
- Coin sparkle effects
- Water effects

Lighting

- Day/Night cycle
- Sunset transitions
- Dynamic lighting effects

Audio System

Create sound effects using JavaScript/Web Audio API.

Include:

- Jungle ambience
- Jump sound
- Coin collection sound
- Damage sound
- Checkpoint sound
- Power-up sound
- Victory sound
- Game over sound

Include mute/unmute controls.

Progression & Rewards

Snake Skins

Unlockable skins such as:

- Emerald Snake
- Golden Snake
- Shadow Snake
- Ancient Temple Snake

Achievement System

Examples:

- First Treasure
- Coin Collector
- Explorer
- Temple Conqueror
- Speed Runner

Display achievement notifications.

Save System

Use localStorage to save:

- High score
- Coins collected
- Unlocked skins
- Achievements
- Highest level reached
- Selected settings

Progress should persist after page refresh.

User Interface

Display:

- Health Bar
- Coin Counter
- Current Level
- Score
- Active Power-Ups
- Pause Menu

Menus should be mobile-friendly and visually polished.

Performance Requirements

The game must:

- Run smoothly on Android devices.
- Run smoothly on iPhones.
- Work in portrait mode.
- Be optimized for mobile browsers.
- Avoid memory leaks.
- Handle touch input reliably.

- Have no console errors.

Deployment Requirements

The final output must:

- Be a single self-contained HTML file.
 - Require no external libraries.
 - Require no installation.
 - Run directly when opened in a browser.
 - Be ready for deployment on Netlify without modification.
 - Be fully playable and free from critical errors.
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Final Output Requirement

Return ONLY the complete HTML code.

Do not provide explanations.

Do not provide summaries.

Do not provide pseudocode.

Generate production-quality code that is immediately playable and deployable on Netlify.

Goal:

Develop a professional-quality mobile game that appeals to teenagers and adults.

Improvements:

- Better visuals and atmosphere.
- Easy to play and more interesting.
- Increased replay value.
- More challenging gameplay.
- Mobile-friendly design.
- Long-term player engagement.

What I Learned

The most important lesson was that high-quality results require multiple prompt iterations. Instead of asking the AI for everything at once, I gradually improved the game by identifying weaknesses in each version and refining the prompt accordingly. The final prompt became much more detailed than the original one, resulting in a significantly better game concept.

Challenges Faced

- Converting a basic snake game into a unique concept.
- Making the game interesting for different age groups.
- Writing detailed prompts without becoming overly complex.
- Ensuring all requested features worked together.

How I Solved Them

- Used iterative prompting.
- Added features in stages.
- Tested ideas and refined requirements.
- Specified gameplay mechanics more clearly.

AI Tools Used:

Claude

Gemini

Chatgpt

Deepseek

Manus

Replit Ai

Game Link

<https://snake-adventure-game-chi.vercel.app/>